Solution Requirements

Date:	21 October 2026	**Team ID:**	Piyush-Balodiya-GenAI-2026
Project Name:	Multimodal Construction Intelligence (CONSTRUCTAI)	**Maximum Marks:**	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform. **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should do). **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

Step 1: Functional Requirements (FR) — Part 1

S.No	Requirement Category	Requirement Description	Priority (H/M/L)
1	**Authentication**	Session-based and JWT token authentication allowing site engineers, EHS officers, and auditors to log in securely with password hashing (bcrypt).	**High**
2	**Authorization Levels**	Role-Based Access Control (RBAC): *Site Inspector* (capture & upload site imagery), *Structural QC Engineer* (verify crack diagnostics & sign off repair sheets), *Project Admin* (manage BOQ materials, weather thresholds, user roster).	**High**
3	**External Interfaces**	Direct HTTPS integration with **Groq AI Cloud API** (LLaMA 3.2 Vision Preview) for sub-second multimodal vision inference, and Open-Meteo Weather API for real-time precipitation radar & wind risk data.	**High**
4	**Transactions Processing**	FastAPI asynchronous request-response pipeline to process multi-megabyte image payloads, base64 encoding validation, audit log persistence, and automated BOQ material calculations.	**High**

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Step 1: Functional Requirements (FR) — Part 2

Remaining functional specifications covering reporting, civil engineering business rules, and safety compliance standards.

S.No	Requirement Category	Requirement Description	Priority (H/M/L)
5	**Reporting**	Automated generation and export of downloadable **Site Safety Audit PDFs**, IS 456 Structural Crack Remediation Blueprints, and BOQ Material Takeoff budget spreadsheets with executive visual charts.	**High**
6	**Business Rules & Logic**	Concrete crack severity logic adheres to **IS 456:2000 Section 35.3.2** (hairline <0.1mm non-aggressive, 0.2mm aggressive, >0.3mm structural risk requiring immediate epoxy/polyurethane pressure grouting). Wind speed >35 km/h triggers mandatory crane stoppage rule.	**High**
7	**Compliance to Regulations**	Strict compliance with **OSHA Safety Regulations (1926.100 PPE)**, **National Building Code of India (NBC 2016)**, and local environmental noise and dust emission safety parameters.	**High**
8	**Other Functional Features**	Multimodal real-time alert trigger (visual red warning banners + audio chime) when un-helmeted workers enter hazardous crane lifting or excavation zones.	**Medium**

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Step 2: Non-Functional Requirements (NFR)

Target performance metrics, security safeguards, system reliability, and accessibility standards for enterprise site deployment.

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	**Performance & Speed**	FastAPI backend handles image ingestion, base64 encoding, and dispatches vision queries with minimum latency.	• API Response < 1.5 seconds • Groq Vision inference < 950 ms
2	**Scalability**	Stateless FastAPI microservices architecture with asynchronous non-blocking worker pools.	• Supports up to 2,000 concurrent site inspection queries without degradation
3	**Security & Data Privacy**	Zero retention of private construction site imagery on external cloud servers. API keys stored via protected environment variables.	• AES-256 encrypted payload • Groq zero-data training guarantee
4	**Reliability & Availability**	Robust offline fallback mode: local cached IS 456 remediation guidelines accessible even when remote cellular network drops.	• 99.8% monthly operational uptime • Graceful fallback handling
5	**Usability & Accessibility**	High-contrast UI designed for outdoor tablets under direct harsh sunlight on construction sites. Intuitive single-click upload flows.	• WCAG 2.1 AA Compliant • Outdoor high-contrast mode supported
6	**Maintainability & Portability**	Modular container-ready codebase with clean separation between FastAPI routes, AI prompt templates, and React UI components.	• Fully portable via Docker • Modular microservice design